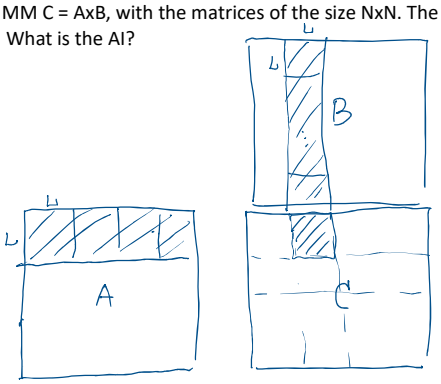
Code Optimization

Tuesday, April 7, 2026 5:31 PM

Tiling

For MM C = AxB, with the matrices of the size NxN. The tile size is LxL. What is the AI?

] Cache size not enough for $2 \cdot N \cdot N \cdot 4B$



$$AI_{\text{w/o tiling}} = \frac{\#\ Ops}{\#\ Bytes} = \frac{N \cdot N \cdot (2N-1)}{N \cdot N \cdot (2N) \cdot 4B + 4 \cdot N^2} = \frac{1}{4}$$

$$AI_{\infty\ cache} = \frac{2N^3}{N^2 \times 4 \times 3} = \frac{1}{6}N \sim O(N)$$

$$AI_{tiling} = \frac{2N^3}{\left((2 \times L \times L) \frac{N}{L} \times 4B + L \times L \times 4B\right) \cdot \frac{N^2}{L^2}}$$

$$= \frac{2N^3}{2L^2 \cdot \frac{N}{L} \cdot \frac{N^2}{L^2} \cdot 4B + L^2 \cdot \frac{N^2}{L^2} \cdot 4B}$$

$$= \frac{2N^3}{\frac{8N^3}{L} + N^2 \cdot 4} = \frac{1}{4}L \quad (N \gg L)$$

$L \uparrow \Rightarrow AI \uparrow$

↓ Cache capacity $\sim L^2 \cdot 3$

- Reuse Distance.

Reuse distance ↑ ⇒ $ size ↑

A B C D A X Y Z

|<——> # unique addrs.

Untiled version

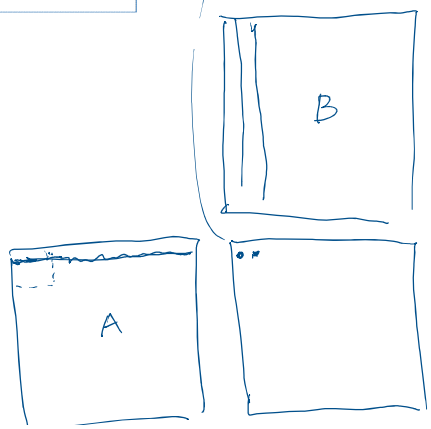
```
for (i = 0; i < N; i++)
   for (j = 0; j < N; j++){
       sum = 0;
       for(k = 0; k < N; k++)
         sum += A[i][k] * B[k][j];
       C[i][j] = sum;
   }
```

&A[0][0]

i=0, j=0 → A[0][0], B[0][0], A[0][1], B[1][0], A[0][2], B[2][0], ...

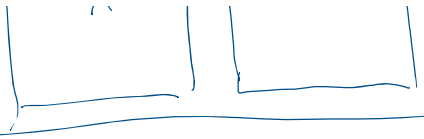
i=0, j=1 → A[0][0], B[0][1],

Reuse distance $\frac{N}{X} * 2$



reuse distance.

tiling L



Memory Perf. (Note tiling shifts AI, making MM more computational intensive)

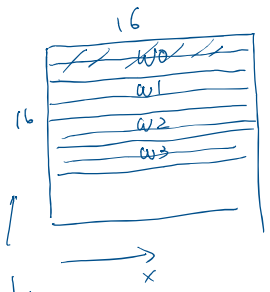
Global memory Acc: A[a + WA * ty + tx] → 2 # line.

TB(0,0) _ W3.

tx: 0~15
ty: 6~7

⇒ A[N·(6~7) + (0~15)]
A[6·N + 0~15]
A[7·N + 0~15]

TB conf: (16, 16)



W0 tx (tidx): 0~15
ty (tidy): 0~1

Shared Mem Access: SA[ty][tx]

// SA[16][16] _ 32 banks.
↓
16 banks

W0. SA[0~1][0~15]

